Title

Human in the Era of Artificial Intelligence and Beyond.

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Abstract

This research examined two dimensions of the human category in the context of technological advancement, with particular emphasis on artificial intelligence (AI). Firstly, it explored how progress in technology may reshape our cognitive representation of the human category. Secondly, it investigated the significance of mental capacities--such as emotional expression and communicative ability--within this categorisation framework. The study specifically aimed to determine whether these capacities are fundamental to human categorisation, as indicated by dehumanisation literature, or if they hold lesser importance, as argued by the causalstatus hypothesis, which posits that such mental abilities, being dependent on biological substrates, may not constitute the core of human identity. Initially, participants responded to baseline items regarding a human subject and a nonhuman AI entity. They were subsequently presented with altered scenarios involving cross-species transplants. These included instances where human subjects depended on AI to perform functional aspects of mental capacities (e.g., emotion and communication), or to replace biological components (e.g., brain and heart). Conversely, nonhuman AI entities were described as possessing either these functional capacities or the corresponding biological structures. The investigation comprised three separate studies. Study 1 involved 55 participants who assessed category membership and typicality. In Study 2, 40 participants evaluated hypothetical scenarios by predicting properties based on fictional attributes ascribed to humans and AI entities, with the aim of identifying whether their predictions for the modified entities aligned with human, AI, or novel categorisations. Study 3 (N = 40) focused on evaluations of category membership, human rights, and associated responsibilities when AI was described as exhibiting complete human functional capacity. Despite the shared functionalities between human and AI entities in these hypothetical contexts, the findings indicated consistency in categorisation, particularly with regard to AI entities. Biological structures were identified as more influential than functional capabilities in defining membership within the human category. These results contribute to a deeper understanding of human conceptualisation and the evolving dynamics of human--AI relations.

Subjects

TECHNOLOGICAL innovations; COMMUNICATIVE competenc intelligence; SELF-expression

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